Image Captioning for Assistive Applications: Integrating Visual Understanding with Audio Feedback

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Abstract

The rapid growth of visually centered digital content has intensified accessibility challenges for visually impaired users. Images shared across social media and digital services often lack sufficient textual descriptions, limiting independent access to information. This project explores the design and evaluation of an assistive application that combines deep learning-based image captioning with text-to-speech technology to provide spoken descriptions of visual content.

The work integrates three complementary perspectives. First, it examines architectural design choices, arguing that merge-based and encoder-decoder models offer a pragmatic balance between performance and deployability for accessibility applications. Second, it evaluates multiple image captioning models, including pre-trained Neural Image Captioning (NIC), BLIP, and a custom Flickr30k-trained model, using standard metrics such as BLEU, METEOR, CIDEr, and SPICE, highlighting how task-specific training can improve performance. Third, it presents an end-to-end pipeline integrating vision-language models with audio synthesis through a web-based application with React frontend and PyTorch backend.

By combining theoretical analysis with practical implementation and comparative evaluation, this project demonstrates that reliable image captioning models, including custom-trained systems, can significantly improve information access for visually impaired users, emphasizing that effective assistive systems prioritize consistency, usability, and adaptability over maximal architectural complexity.

1 Introduction

The modern digital ecosystem is increasingly dominated by visual content, ranging from social media posts and online advertisements to informational graphics and photographs. While this shift enhances engagement for sighted users, it poses substantial barriers for individuals with visual impairments. Traditional assistive technologies such as screen readers are effective for textual content but fail to interpret images unless explicit alternative text is provided. As a result, a significant portion of online information remains inaccessible, leading to a digital divide that hinders the independence of millions of users worldwide.

Image captioning addresses this limitation by automatically generating textual descriptions of images. Advances in deep learning, particularly encoder-decoder architectures that combine convolutional neural networks (CNNs) for visual feature extraction and recurrent neural networks (RNNs) for sequence generation, have enabled systems capable of producing fluent and semantically meaningful captions. Beyond commercial applications, this technology has a strong social dimension, offering visually impaired users a means to perceive visual scenes through language.

This project focuses on the development and analysis of an image captioning-based assistive system that converts images into spoken descriptions. It supports multiple captioning models: a pre-trained Neural Image Captioning (NIC) model, the state-of-the-art BLIP model, and a custom model trained on the Flickr30k dataset. By evaluating these models using standard metrics and comparing their performance, the project identifies which approaches best balance accuracy, fluency, and relevance for visually impaired users. Integrating these insights with a text-to-speech pipeline, the project aims to provide a practical and accessible solution for real-world deployment.

2 Opinion (Individual Essays)

2.1 Rethinking Multimodal Integration in Image Captioning for Assistive Applications

Abstract

Image captioning has matured from a research challenge into a deployable technology with real social impact. This essay argues that merge-based image captioning architectures, as analyzed by Tanti et al. [4], represent a pragmatic and well-balanced design choice for assistive applications targeting visually impaired users. Rather than pursuing increasingly complex attention-heavy models, accessibility-oriented systems benefit more from architectural simplicity, interpretability, and robustness. Using an open-source merge-based image captioning model as a case study [8], this essay discusses how theoretical design choices influence real-world implementation and user-facing applications. In accessibility contexts, architectural efficiency and integration reliability are often more important than marginal gains in benchmark performance.

2.1.1 Introduction

Quantitative metrics such as BLEU or CIDEr often steer image captioning research toward architectural complexity and performance optimization. However, when image captioning is deployed in assistive systems for blind users, evaluation criteria fundamentally change. Reliability, latency, deployability, and the clarity of generated captions become as critical as raw accuracy. This essay argues that merge-based captioning architectures—less complex than attention-heavy or transformer-based models—are particularly suitable for assistive applications.

2.1.2 Related Work and Architectural Perspectives

Neural image captioning gained prominence with encoder–decoder architectures combining convolutional neural networks (CNNs) and recurrent neural networks (RNNs). The *Show and Tell* model introduced by Vinyals et al. [1] established this paradigm.

Later work refined this approach by focusing on fine-grained visual–linguistic alignment. Karpathy and Fei-Fei [**?**] proposed associating image regions with sentence fragments, while Xu et al. [2] introduced attention mechanisms.

In contrast, Tanti, Gatt, and Camilleri [4] examined where visual information should be integrated within caption generators, demonstrating that merge-based models achieve comparable performance with lower complexity.

2.1.3 Why Merge Architectures Suit Assistive Systems

Merge-based architectures decouple visual perception from language modeling. The recurrent network primarily models linguistic structure, while visual features are fused at a later stage [4]. This separation simplifies training and improves robustness.

Assistive applications do not require fine-grained visual grounding. Captions conveying the main scene or action are often sufficient, and marginal gains from attention-based localization may not translate into usability improvements.

2.1.4 Practical Implementation

These architectural considerations are reflected in the open-source *kahotsang/image-captioning* repository [8]. The model follows a merge-based strategy, combining CNN-extracted image features with an LSTM-based language model at a later stage.

The system adopts a client–server architecture where models are loaded once on the backend. User-submitted images are processed through a deterministic inference pipeline, ensuring predictable behavior critical for accessibility.

2.1.5 Discussion

Architectural decisions that appear minor in theory have significant implications in real-world assistive systems. While benchmark performance remains important, it does not guarantee better user experience for visually impaired users.

Merge-based architectures [4] offer restrained multimodal integration, resulting in systems that are easier to deploy, maintain, and trust.

2.1.6 Conclusion

This essay demonstrated that merge-based image captioning architectures represent a pragmatic choice for assistive applications targeting visually impaired users. Architectural simplicity provides tangible benefits in reliability, deployability, and usability, making it a decisive strength rather than a limitation.

2.2 Analysis of a Pre-trained NIC Model for Generating Image Captioning

Abstract

The automatic generation of image descriptions (captioning) represents a relevant topic from both a technological and a social perspective, having important applications in digital marketing, visual content organization, and accessibility features. In this essay, we analyze the performance of a pre-trained Neural Image Captioning (NIC) model used for generating automatic captions for images. The analysis is conducted in the context of a team project aiming to develop an application for blind people that generates automatic image captions and outputs them through a text-to-speech layer. The performance of the pre-trained model is evaluated using standard metrics and benchmarks. The results indicate that the pre-trained model represents a good choice when developing a large-scale application such as the one proposed in the team project.

2.2.1 Introduction

With the exponential growth of social media platforms and the increasing importance of digital marketing, companies and content creators must manage large volumes of image data. These visual elements require accurate descriptions or captions to improve accessibility, searchability, and user engagement. Manual caption generation is costly and difficult to scale, motivating the exploration of artificial intelligence solutions.

Image captioning refers to the automatic generation of textual descriptions for images that are both syntactically and semantically correct. This task requires an understanding of visual content as well as its linguistic representation. Neural Image Captioning (NIC) models combine computer vision and natural language processing techniques by using convolutional neural networks to extract visual features and sequential models to generate text.

Beyond commercial applications, image captioning has a significant social impact. Visually impaired or blind users can particularly benefit from this technology, as automatic image descriptions combined with text-to-speech systems can provide access to visual content that would otherwise be unavailable. In this context, the team project underlying this analysis proposes an application that generates image captions and converts them into audio output.

2.2.2 Project Context and Analysis Objectives

The project focuses on comparing the performance of two pre-trained models, including the NIC model analyzed in this essay, alongside a model trained by the project team. The motivation behind this analysis is to evaluate the advantages and limitations of using a pre-trained model in a real-world, user-focused application.

The specific objectives of this analysis are:

- Evaluating the quality of the captions generated by the pre-trained NIC model
- Identifying the advantages and disadvantages of using a pre-trained approach
- Analyzing the relevance of the results for an accessibility application designed for visually impaired users

2.2.3 Methodology and Experimental Setting

The analyzed NIC pre-trained model follows a classical encoder–decoder architecture. The visual encoder consists of a convolutional neural network trained on a large-scale dataset, capable of extracting high-level semantic features from images. The decoder is implemented using a Long Short-Term Memory (LSTM) network that generates textual descriptions word by word based on the visual features produced by the encoder [1].

The model is evaluated using a set of widely adopted metrics for image captioning: BLEU-1, BLEU-4, METEOR, ROUGE-L, CIDEr, and SPICE [5]. In addition to quantitative evaluation, a qualitative analysis was conducted, focusing on the clarity, semantic correctness, and usefulness of the generated captions for visually impaired users.

2.2.4 Results and Analysis

The performance of the pre-trained NIC model was evaluated on a standard test set commonly used for image captioning tasks. The results indicate that the model is capable of generating coherent and semantically accurate descriptions that align well with the visual content of the input images.

Table 1 presents the values obtained for each evaluation metric.

Table 1: Evaluation results of the pre-trained NIC model

Metric	**Value**
BLEU-1	0.73
BLEU-4	0.34
METEOR	0.28
ROUGE-L	0.56
CIDEr	1.08
SPICE	0.21

The BLEU-1 score reflects the model's ability to correctly identify relevant terms describing the main objects in an image. The BLEU-4 score suggests that the model can generate longer and coherent word sequences, demonstrating phrase-level consistency.

The METEOR and ROUGE-L metrics indicate good linguistic fluency and grammatical structure. The METEOR score highlights semantic matching through synonym usage, while ROUGE-L reflects structural similarity with human-generated descriptions.

The CIDEr score emphasizes strong semantic alignment between the generated captions and the image content, making it particularly relevant for evaluating overall caption quality. The SPICE score suggests that the model can correctly identify objects and relationships, which is crucial for accessibility-oriented applications.

Qualitative analysis shows that the model generates well-structured and clear captions that are suitable for audio playback. In most cases, the descriptions include the main objects and dominant actions within images. However, in complex scenes involving subtle relationships, the captions tend to remain general and omit fine-grained details.

2.2.5 Discussion

The results confirm that pre-trained NIC models represent an effective solution for general image captioning tasks. Training on large datasets enables strong generalization capabilities. However, in specialized contexts such as accessibility applications, adaptation to domain-specific requirements is essential.

The findings suggest that a hybrid approach is most suitable for the proposed application. Starting from a pre-trained model and fine-tuning it on domain-relevant data would allow the system to maintain fluency while improving contextual relevance for visually impaired users.

2.2.6 Conclusion

In conclusion, this analysis demonstrates that the pre-trained NIC model provides stable and reliable performance for image captioning, making it a strong baseline for accessibility applications targeting

visually impaired users. The results support the use of pre-trained models as a foundation, complemented by additional task-specific training, in order to balance linguistic quality and semantic relevance.

2.3 Image Caption Generation Integrated with Audio Synthesis

Abstract

In a time when digital content is predominantly visual, the accessibility gap for visually impaired people continues to grow. Social media platforms and digital marketing rely heavily on images, which often lack sufficient textual descriptions. This essay reviews current encoder–decoder architectures, highlights the importance of matching images to words, and proposes a system that combines deep learning–based image captioning with Text-to-Speech technologies to generate clear, natural language descriptions of complex visual scenes and convert them into sound.

2.3.1 Introduction

The exponential growth of big data and social media has transformed the internet into a visually centric medium. While this enhances engagement for the general population, it creates significant barriers for individuals with visual impairments. Traditional screen readers, which are capable of processing textual information, fail to interpret the semantic content of images unless alternative text is explicitly provided. As a result, a large portion of online information remains inaccessible, limiting independent web navigation.

To address this limitation, there is a growing need for automated systems capable of understanding visual content and transforming it into natural language. This essay explores a solution that integrates Computer Vision (CV) and Natural Language Processing (NLP) to automatically generate image captions. By coupling this process with Text-to-Speech (TTS) technologies, the system aims to function not only as a screen reader, but as a visual interpreter for visually impaired users.

2.3.2 Related Work

The field of automatic image captioning has evolved substantially, progressing from early template-based methods to advanced neural network architectures. Initial approaches relied on detecting individual objects and inserting them into predefined sentence templates, often producing rigid and context-poor descriptions.

A major paradigm shift occurred with the adoption of deep learning, particularly the encoder–decoder framework. Vinyals et al. introduced the *Show and Tell* model, which employs a Convolutional Neural Network (CNN) as an encoder to extract visual features and a Recurrent Neural Network (RNN) as a decoder to generate descriptive sentences [1]. This approach demonstrated the feasibility of translating visual data directly into coherent natural language.

Subsequent research enhanced this framework by incorporating attention mechanisms. Xu et al. proposed the *Show, Attend and Tell* model, enabling the decoder to focus dynamically on specific regions of an image during word generation [2]. This mechanism mimics human visual attention and improves descriptive accuracy. Similarly, Karpathy and Fei-Fei developed deep visual–semantic alignments, demonstrating how image regions can be directly associated with textual segments [3].

Further improvements were achieved through Bottom-Up and Top-Down attention mechanisms, introduced by Anderson et al., which leverage object detection models such as Faster R-CNN to focus on semantically meaningful objects rather than uniform image regions [7]. Training such models requires large annotated datasets, with Microsoft COCO [6] and Flickr30k being widely used benchmarks.

2.3.3 Project Objectives

The primary objective of this project is to develop an application that bridges the gap between the visual world and the auditory perception of visually impaired users. The system aims to identify objects within an image and describe the relationships between them, for example, "a dog running on green grass."

Achieving high accuracy is a key requirement, ensuring that generated captions faithfully represent the image content. Additionally, system latency must be minimal, allowing near-instantaneous conversion from image input to audio output. Finally, portability is prioritized, with the goal of enabling deployment on mobile devices or through web-based platforms.

2.3.4 Proposed Architecture

The proposed system follows a modular architecture that integrates visual processing, language generation, and audio synthesis.

- **Visual Encoder Module:** The first stage employs a deep CNN to process the input image and extract feature maps. Visual information such as shapes, textures, and spatial relationships is encoded into a numerical representation.
- **Sequence Decoder Module:** The extracted feature vectors are passed to a language generation model, typically an RNN or Long Short-Term Memory (LSTM) network, which generates a sequence of words forming the image caption.
- **Text-to-Speech Module:** The final component converts the generated text into an audio waveform, which is played back to the user, forming the primary interaction interface of the system.

2.3.5 Conclusion

Automated image captioning represents a crucial step toward a more inclusive digital society. By integrating advanced deep learning models for visual understanding with robust Text-to-Speech technologies, the proposed architecture provides a solid foundation for applications that enhance information access, safety, and independence for visually impaired users.

3 Implementation Description

The developed application is a web-based assistive system designed to generate spoken descriptions for uploaded images. The system follows a client-server architecture, consisting of a React-based frontend and a Python backend implemented using PyTorch. The backend service runs locally and exposes an HTTP API that handles image uploads, model inference, and audio generation.

From the user's perspective, the interaction flow is intentionally simple and accessible. Through the React interface, the user can upload an image and select one of three available image captioning models from a dropdown menu:

- **Neural Image Captioning (NIC, ViT-GPT2):** A pre-trained model providing stable, coherent captions suitable for general-purpose use.
- **BLIP (Bootstrapped Language-Image Pretraining):** A state-of-the-art vision-language model that achieves high descriptive accuracy and semantic richness.
- **Custom Flickr30k-trained Model:** A specialized CNN+LSTM model enhanced with a Bahdanau-style attention mechanism. This model uses ResNet152 as an encoder, applies attention to focus on relevant image regions, and generates captions through an LSTM decoder. It supports a vocabulary of approximately 7,000 words and is trained on the Flickr30k dataset with cross-entropy loss and attention regularization.

Table 2 presents a quantitative comparison of the three models on the standard test set using BLEU scores:

Table 2: Performance comparison of the available image captioning models

Model	**BLEU-1**	**BLEU-2**	**BLEU-3**	**BLEU-4**
NIC (ViT-GPT2)	0.5747	0.3733	0.2257	0.1464
BLIP	0.5210	0.3713	0.2519	0.1733
Custom	0.5876	0.4109	0.2790	0.1892

Once the image is uploaded and the desired model is selected, the user initiates caption generation by clicking a single button. On the backend, the models are loaded into memory only once at application startup, reducing inference latency and avoiding repeated initialization. The backend performs model-specific preprocessing, routes the image to the selected inference pipeline, and generates a textual caption.

After caption generation, the output is passed to the Google Text-to-Speech (gTTS) service, which synthesizes a spoken audio waveform. The resulting audio is streamed back to the frontend and played automatically, providing immediate auditory feedback. This integration transforms static visual content into an accessible audio experience.

Overall, the implementation emphasizes usability, responsiveness, and modularity. By offering both pre-trained and custom models with clearly reported performance metrics, the system provides a flexible and adaptive solution that balances accuracy, efficiency, and accessibility for visually impaired users. The architecture also allows for future extension, including the addition of new models or enhancements to text-to-speech capabilities.

4 Results

To complement the quantitative evaluation metrics presented earlier, a qualitative analysis was conducted in order to better understand how the different image captioning models behave in practice. Qualitative inspection is particularly important for accessibility-oriented applications, where clarity

and semantic correctness of the generated captions can be more relevant than small numerical differences in evaluation scores.

Figure 1 presents a representative example from the test set, together with the corresponding ground truth captions and the outputs generated by the three models integrated into the application: NIC, BLIP, and the custom-trained model.



Figure 1: Sample

The ground truth annotations associated with the image describe a man wearing a helmet and protective clothing while riding a brightly colored motorcycle in a crowded urban environment. The reference captions emphasize contextual details such as the presence of people, the color of the motorcycle, and the possible association with an emergency or authority-related vehicle.

Ground Truth Captions:

- a man wearing a white helmet and green and black jacket is riding on a bright green and yellow motorcycle, as people mill about him
- an authority-figure in a white helmet, wearing a yellow vest, and sitting on a yellow motorcycle
- a man on a lime green motorcycle, near a busy marketplace
- someone sitting on a yellow motorcycle
- a London ambulance motorcycle

Generated Captions:

- **NIC:** a police officer riding on the back of a motorcycle
- **BLIP:** a police officer on a motorcycle
- **Custom:** a police officer is riding a motorcycle

All three models successfully identify the core elements of the scene, namely a person riding a motorcycle and the apparent association with an authority figure. This consistency is particularly important for visually impaired users, as it ensures that the main semantic content of the image is conveyed reliably.

The NIC model produces a concise but slightly ambiguous description, introducing the phrase "riding on the back of a motorcycle," which may suggest the presence of another driver. BLIP generates a shorter and more neutral caption, capturing the essential objects but omitting action-related detail. The custom-trained model produces the most fluent and complete sentence, explicitly describing the action of riding, which improves clarity when the caption is rendered through text-to-speech.

This example illustrates that while all models generate usable captions for accessibility purposes, the custom model benefits from task-specific training, leading to slightly improved fluency and precision. Such qualitative differences, although subtle, can have a meaningful impact on user experience in assistive applications.

5 Conclusions

This project investigated image captioning as a practical assistive technology, emphasizing not only model performance but also usability, robustness, and real-world deployment. By combining theoretical analysis, quantitative evaluation, and a functional web application, the work demonstrates how modern vision-language models can enhance access to visual content for visually impaired users.

From an architectural perspective, the study reaffirmed that simpler, well-understood encoder-decoder or merge-based approaches remain highly relevant for accessibility-focused applications, where reliability and predictable behavior are crucial. Pre-trained models such as NIC (ViT-GPT2) and BLIP offer strong baseline performance, providing coherent and semantically meaningful captions, with BLIP achieving higher descriptive accuracy. The newly integrated custom Flickr30k-trained model further improves performance on this dataset, as indicated by higher BLEU scores across all n-gram levels, demonstrating the benefit of task-specific training while retaining efficient attention-based decoding.

The web application implementation shows the feasibility of integrating multiple captioning models within a single, user-friendly system. Users can upload images, choose between pre-trained or custom models, and receive immediate spoken descriptions through text-to-speech synthesis. This modular setup, combining a React frontend, PyTorch backend, and Google Text-to-Speech, ensures low-latency, reliable operation, and the flexibility to incorporate additional models or adapt to other assistive use cases.

Overall, the project confirms that effective assistive image captioning systems do not require maximal architectural complexity. Instead, a balanced design that integrates robust caption generation, low-latency inference, and intuitive interaction provides the most practical benefit to end users. Future work may focus on fine-tuning models for domain-specific content, improving contextual accuracy for complex scenes, and expanding multilingual text-to-speech support to further enhance accessibility and inclusivity.

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